

Smart Flashcard Learning System

Built using C++ Programming

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Objectives of the Project



Enhance Learning Efficiency

Utilize active recall to strengthen neurological pathways for better data retention.



Implement Spaced Repetition

Show cards at specific intervals based on difficulty to maximize memory efficiency.



Personalized Adaptive Learning

System automatically prioritizes difficult concepts for more focused study sessions.

 Abstract brain connections

Limitations vs Advantages

✖ Existing System (Manual)

- Information Overload
- Ineffective Review Strategy
- Rapid Forgetting of complex data
- No progress tracking mechanism

✔ Proposed System (Digital)

- + Adaptive Learning Algorithms
- + Better Information Retention
- + Smart Progress Tracking & Analytics
- + Efficient storage & instant access

Smart Flash Cards Learning Code

```
#include
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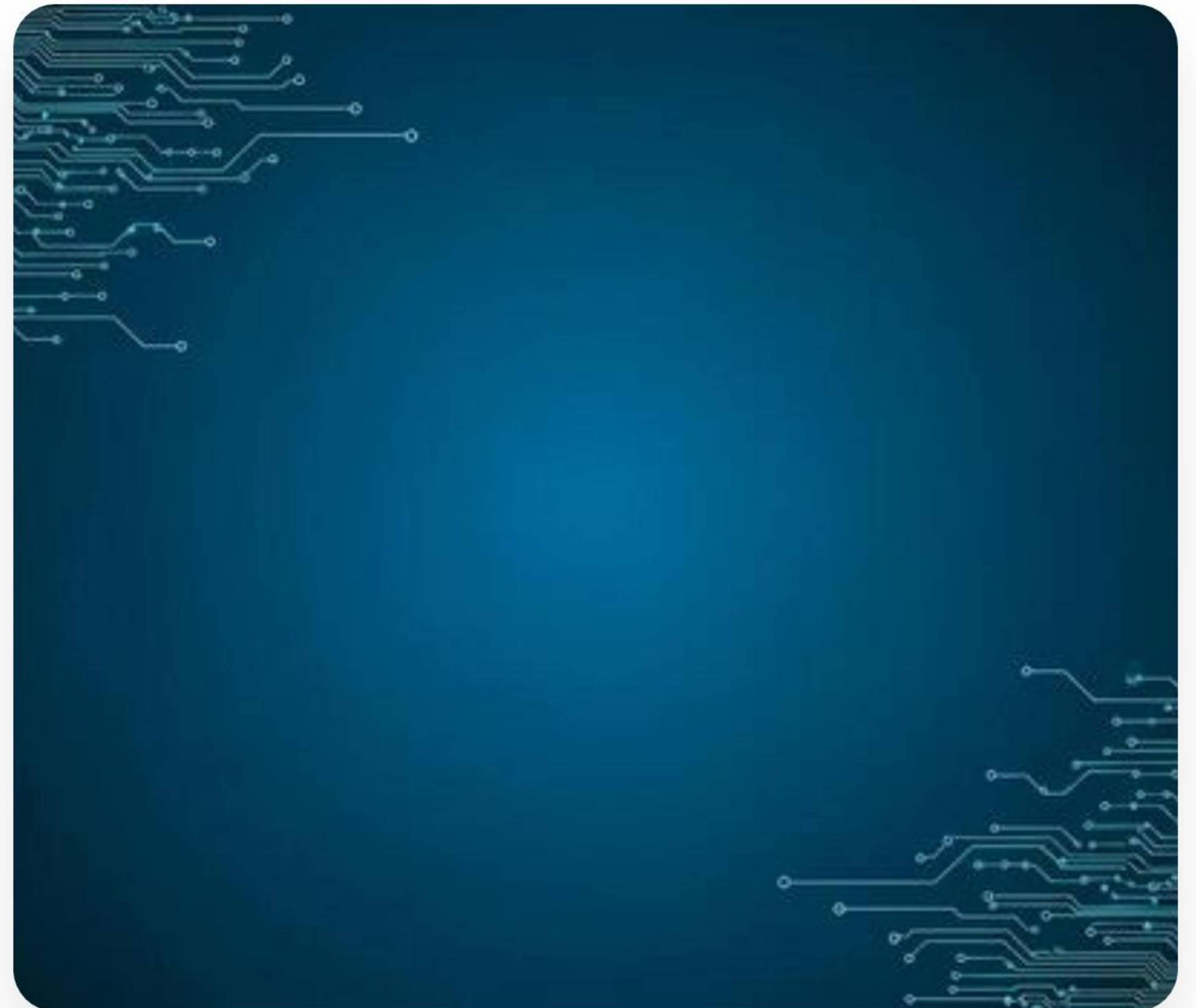
using namespace std;

struct FlashCard {
    string question;
    string answer;
    int correctCount;

    FlashCard(string q, string a)
        : question(q), answer(a), correctCount(0) {}
};

class FlashCardSystem {
private:
    vector cards;

public:
    void addCard() {
```



Smart Revision Logic



```
void studyCards() {  
    // Sort cards by correctCount (ascending)  
    // Smart: Study what you forget most!  
    sort(cards.begin(), cards.end(),  
        [](const FlashCard& a, const FlashCard& b) {  
            return a.correctCount < b.correctCount;  
        });  
  
    for (auto& card : cards) {  
        string userAnswer;  
        cout << "\nQuestion: " << card.question;  
        cout << "\nYour Answer: ";  
        getline(cin, userAnswer);  
  
        if (userAnswer == card.answer) {  
            cout << "Correct!\n";  
            card.correctCount++;  
        } else {  
            cout << "Wrong! Ans: " << card.answer;  
        }  
    }  
}
```

Conclusion & Future Scope

Modern Learning Solution

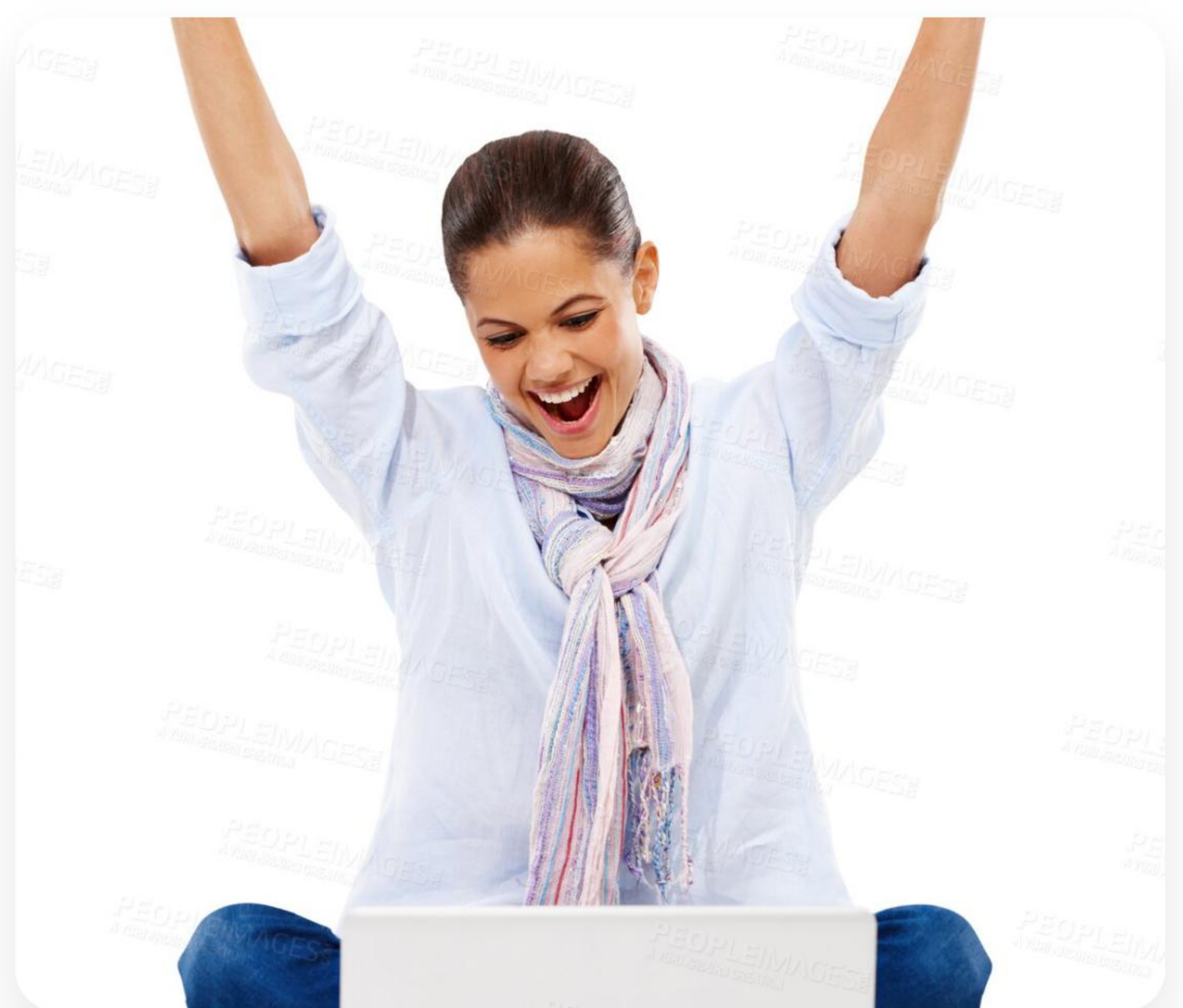
A digital approach to one of the most powerful study techniques ever discovered.

Improves Retention & Efficiency

Significantly reduces study time while increasing long-term data recall.

Relevant for Digital Education

Ready for integration into modern e-learning platforms for 21st-century students.





Thank You!

Smarter Study Approach | Ready for Future Success

Questions? Reach us at PEC (A), Surampalem

Image Sources



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